STEAMROOT · DESIGN DECISION RECORD

SteamRoot recommendations accepted

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DOCUMENT CONTROL

Document ID	STR-DDR-002	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/steamroot		
Source	docs/decisions/0002-recommendations-accepted.md @ 276b380		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Item 1 consequence updated: evaporator bank now in the model within R7 (STR-CAL-001 v0.4)
0.2	2026-10-02	Amish Chadha	Items 10 to 12 decided by Amish on 2026-10-02 (STR-DEC-001, items 1 to 3); the ruling is still to be obtained
0.1	2026-09-25	Amish Chadha	Record the TRL 3 review recommendations accepted by Amish, what changed in the repo, and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 9); items 10 to 12 remained open at this record and were decided by Amish on 2026-10-02 (STR-DEC-001, items 1 to 3): "i approve your recommendations for all 555 open decisions." The ruling of item 10 is still to be obtained.

Context

The TRL 3 review note (`docs/REVIEW.md`, session 2026-09-25) and STR-DDR-001 left several items "Proposed, awaiting Amish", most with a recommendation. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every open item that carried a recommendation is therefore decided as recommended. Where the recommendation listed several options, the recommended option is the decision. Items with no recommendation stay open. TRL 4 remains on hold by Amish's earlier instruction, and the repo stays at `trl: 3`, `trl_target: 3`.

> **Safety:** None of these decisions replaces the written ruling from the local boiler authority, which is still an open prerequisite for any detail design or build.

Decision

Table 1. Items decided by Amish on 2026-09-25.

#	Item	Decision	What changed in the repo
1	R5 efficiency and R4 wood use	Decided by Amish, 2026-09-25: go with recommendation. Keep the targets (65 %, 4 kg/m²) and study a convective evaporator bank in the flue with controlled primary and secondary air at the next paper step.	STR-CAL-001 v0.2, section 4, adds the study: about 3.3 m of 25.4 mm tube (0.26 m²) reaches 65 % at λ = 2.0; almost none is needed at λ = 1.5. R4 needs about 72 %. Baseline design unchanged; R4 and R5 still not met.
2	R2 treatment rate	Decided by Amish, 2026-09-25: go with recommendation (a). Accept about 1.9 m²/h at 15 cm and restate the 5 cm target at about 3.4 m²/h, until soil tests exist.	STR-REQ-001 v0.4: R2 from 2 and 5 m²/h to 1.85 (about 1.9) and 3.4 m²/h. Status from not met to at risk (1.88 and 3.41 m²/h).
3	R6 pressure	Decided by Amish, 2026-09-25: go with recommendation. 0.1 bar gauge at the header, where the vent limit applies, and 0.15 bar gauge at the coil inlet.	STR-REQ-001 v0.4: R6 split. Status from not met (0.110 bar against 0.1) to met (header 0.033, inlet 0.110). Drawing note updated.
4	R9 relief valve	Decided by Amish, 2026-09-25: go with recommendation (both parts). Word the set pressure as "15 psi (1.03 bar) or less" and size the valve for the dry-coil refeed flash (about 155 kg/h, 14 mm orifice) as well as 30 kg/h.	STR-REQ-001 v0.4 reworded. Maker's capacity table checked: Watts Series 315, 3/4 in, 15 psi, 375 lb/h (170 kg/h), so the BOM valve is kept. Status from at risk to met on paper. BOM note, drawing note and CAL-001 updated.
5	R11 mass	Decided by Amish, 2026-09-25: go with recommendation (a). "500 kg or less as towed, with the tank drained"; fill on site.	STR-REQ-001 v0.4 restated. Status from not met (536 kg full) to met (402 kg as towed). Width stays at risk (1.48 m).
6	R12 budget	Decided by Amish, 2026-09-25: go with recommendation (a). Raise the budget to about $2,200 for about 15 % contingency.	`project.yaml` `budget_usd` from 1800 to 2200; R12 from $1,800 to $2,200. Status from not met to met ($1,915, $285 contingency).
7	Firebox lining	Decided by Amish, 2026-09-25: go with recommendation. Ceramic fiber lining, not castable (castable would add about 200 kg).	Recorded in STR-PRC-001 v0.4. Design already used fiber; no geometry change.
8	Hose between hoods	Decided by Amish, 2026-09-25: go with recommendation. One hose moved between hoods with a steam-rated coupling, after steam is diverted to the vent.	Recorded in STR-PRC-001 v0.4. BOM already had one hose; no change.
9	Fuel	Decided by Amish, 2026-09-25: go with recommendation (the design proposal). Wood only, no liquid fuel backup.	Recorded in STR-PRC-001 v0.4 and STR-DDR-001 v0.2.

Items still open

Table 2. Items left open at this record; decided on 2026-10-02.

#	Item	Status
10	Written ruling from the local boiler authority	Approved as a prerequisite; not obtained. Blocks detail design and any build. Confirmed by Amish on 2026-10-02 (STR-DEC-001, item 1): the design is reviewed against the ruling and anything stricter is adopted.
11	Which jurisdiction is "home" and which authority to ask	Decided by Amish, 2026-10-02 (STR-DEC-001, item 2): Texas; the ruling is asked of the Texas Department of Licensing and Regulation's boiler program.
12	The named market garden and nursery (first co-design partners)	Decided by Amish, 2026-10-02 (STR-DEC-001, item 3): one market garden and one nursery within driving distance of Irving that already steam, solarize or chemically treat soil, found through Texas A&M AgriLife Extension; none is named or agreed yet.

Consequences

- Requirement status (STR-CAL-001 v0.2, 15 checks): 2 not met (R4, R5), 3 at risk (R2 at 15 cm and at 5 cm, R11 width), 1 not verifiable at TRL 3 (R1), 9 met. Before these decisions: 7 not met, 2 at risk, 1 not verifiable, 4 met.
- Documents bumped: STR-PRB-001 v0.4, STR-PRC-001 v0.4, STR-REQ-001 v0.4, STR-CAL-001 v0.2, STR-DDR-001 v0.2. Drawing STR-DWG-002 moved to Rev P2 for note changes (relief valve rating, towed mass, R6 limits); geometry is unchanged.
- The evaporator bank was added to the model on 2026-10-02: 6.4 m of 15.88 mm tube with one coil turn fewer keeps the heated section at 7.3 L, inside R7, with R6 met and R11 mass at risk (STR-CAL-001 v0.4).
- TRL 4 work that these decisions point to (measuring efficiency, soil tests that would settle R2, confirming the valve capacity on a bought unit) is decided but on hold, because TRL 4 is on hold by Amish's instruction.
- Cross-repo actions: none.